

What the score hides

A bioinformatics-agent benchmark scores each question with a single pooled number. I replicated the runs, replicated the grading, and re-derived every answer key to see what that number leaves out.

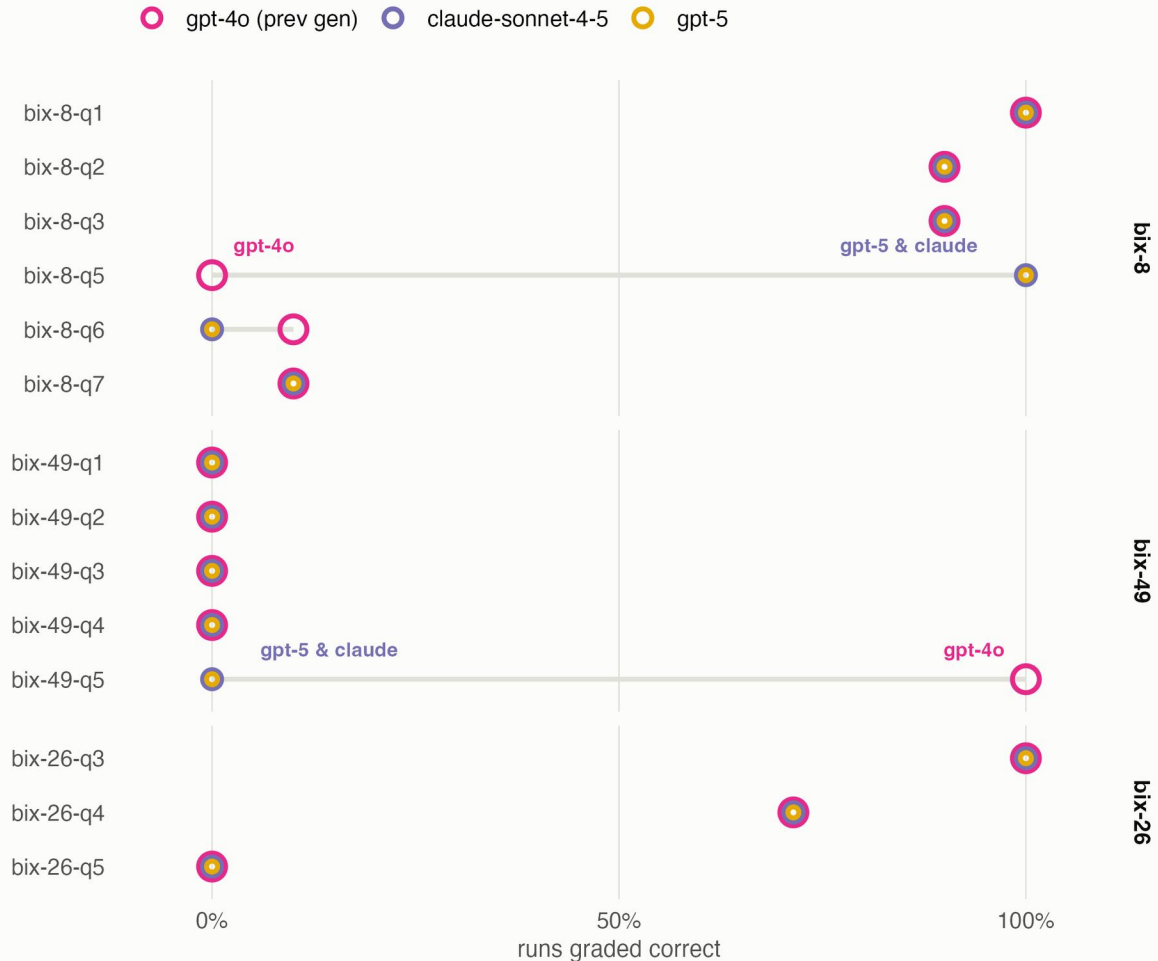
140	agent runs, 10 per question
3,900	grading verdicts, 3 graders × 10 rounds
14	answer keys re-derived from raw data

swipe →

FIGURE 1

Grader disagreement follows model generation, not model family

Share of answered runs graded correct per question, by grader (majority of 10 grading rounds per answer). Nested rings = the graders agree on that question.



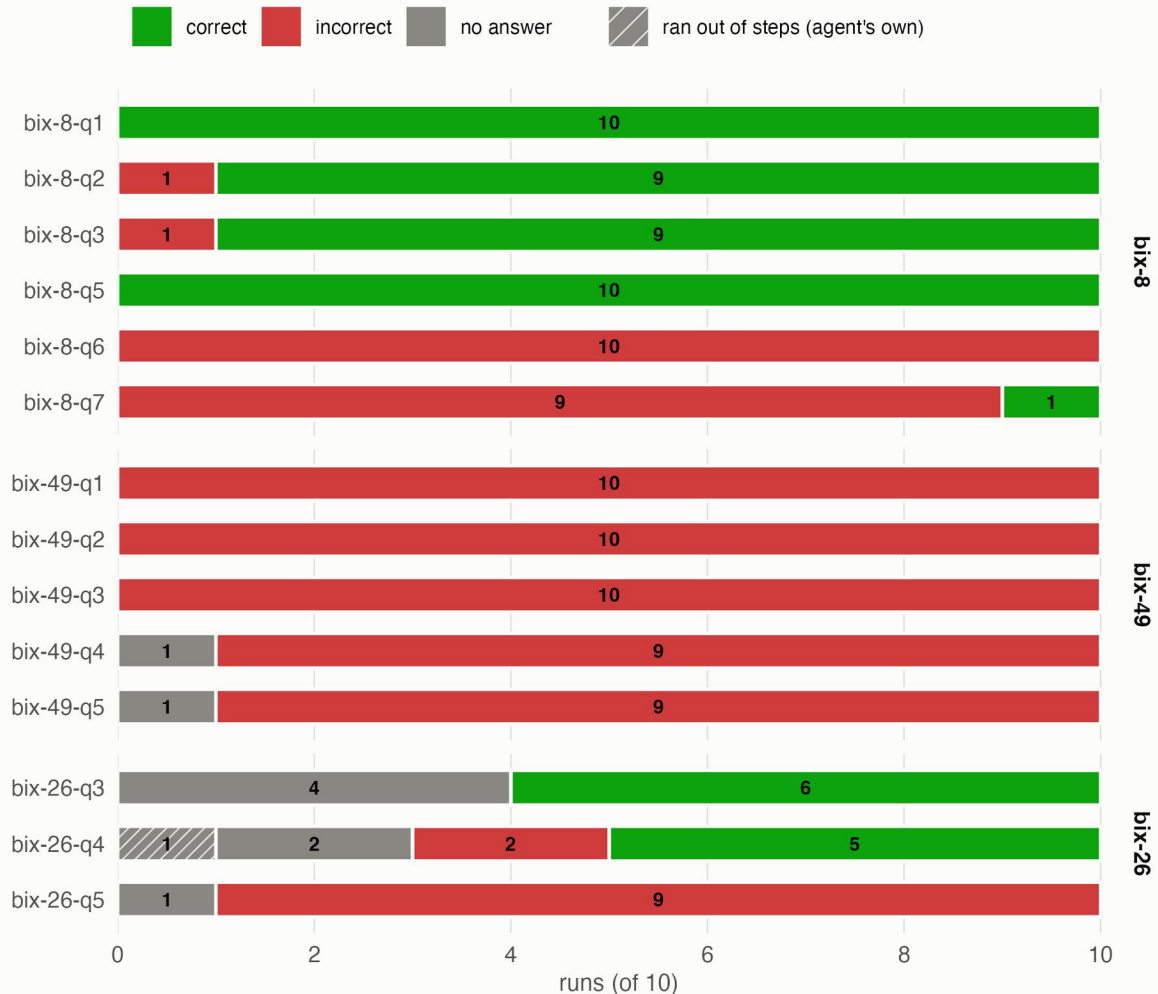
Same 130 answers under all three graders. gpt-5 is the primary grader for the remaining figures.

Two current-generation graders agree on all 130 answers. The shipped previous-generation grader contradicts itself on 14.6% of them.

FIGURE 2

No answer is its own outcome, not a wrong answer

10 replicate runs per question (claude-sonnet-4-5, temperature 1.0); correctness by gpt-5 majority vote.



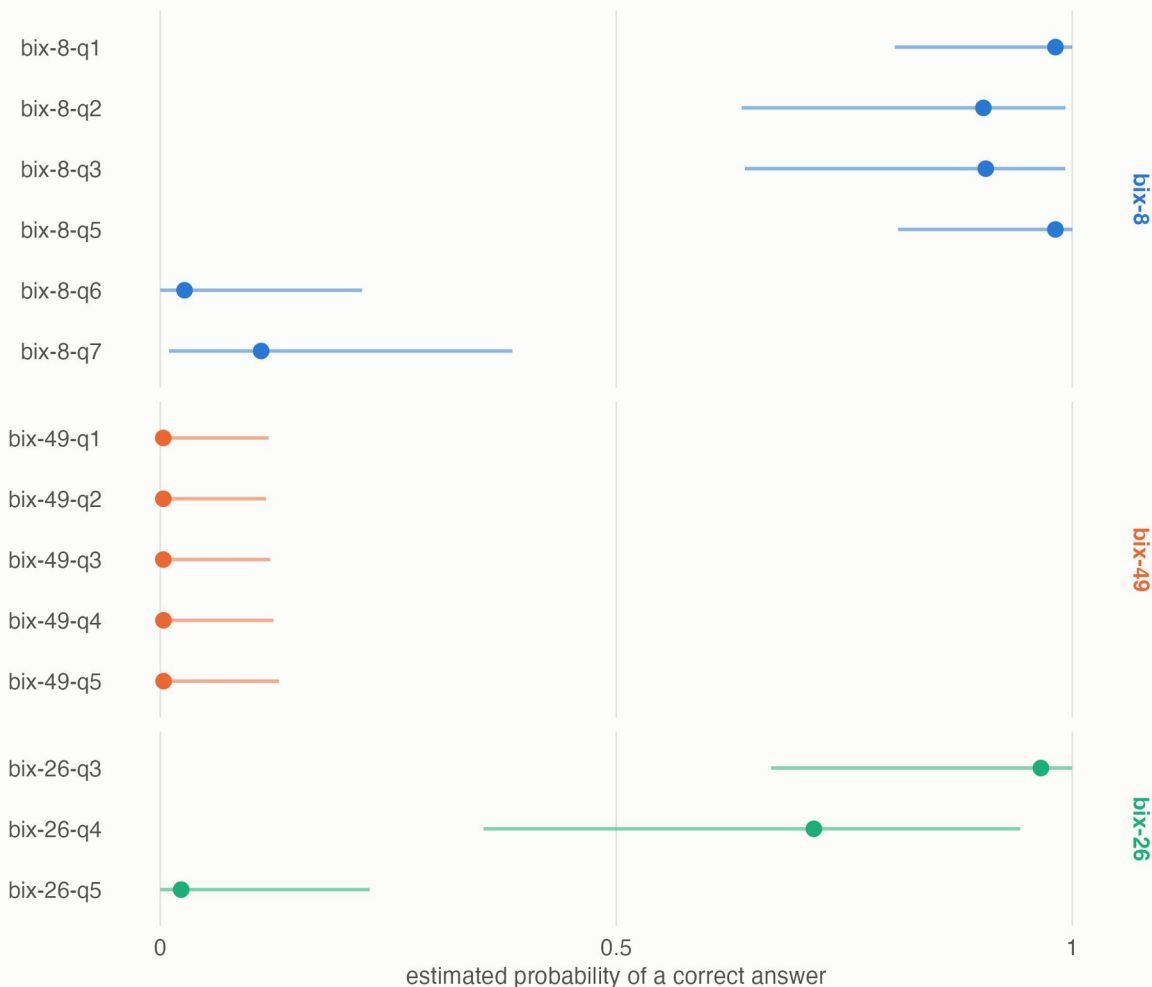
BixBench: 3 capsules covering 14 questions, 10 replicate runs each = 140 runs.

10 of 140 runs never answered - and 9 of those died following the harness's own package-install instruction.

FIGURE 3

Given an answer, correctness is a property of the question, not luck

Answered runs only. Dot = a question's estimated probability of a correct answer (posterior median); band = 95% uncertainty about that probability.



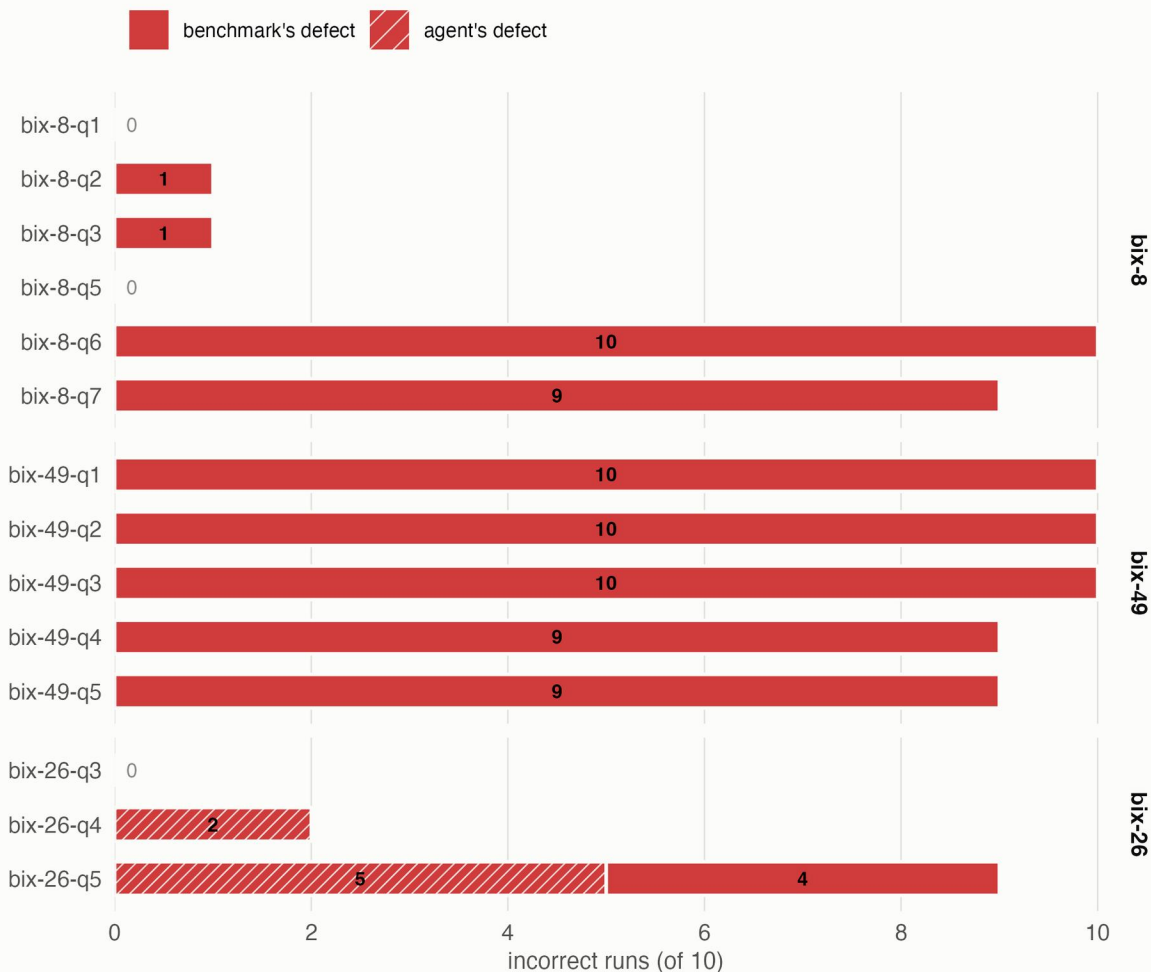
Bernoulli GLMMs, question nested in capsule; gpt-5 majority-vote grading. Question-level SD (stage 2): 3.2 log-odds.

Per-question success is near-certain or near-impossible, with almost nothing between: the outcome belongs to the question.

FIGURE 4

Most incorrect answers trace to the benchmark

Every incorrect run attributed by re-deriving both the answer key and the agent's answer from the raw capsule data (env/REVIEW_REPORT.md).



gpt-5 majority-vote grading. bix-8-q5's print-floor key is absent only because gpt-5 grades $p = 8.1e-194$ as satisfying " $p < 2.2e-16$ ".

Re-deriving every key from the raw data: most incorrect answers trace to the answer key or the question's wording, each cause verified.

READ THE FULL AUDIT

Every claim is recomputable

The writeup, the code, the grading data, and the case-by-case failure attribution are public - every recomputation is a committed script.

Writeup + code

github.com/Anqi-Dai/BixBench_eval

Interactive run audit

anqi-dai.github.io/BixBench_eval/results/figures/fig4_interactive.html

Data DOI

[10.5281/zenodo.22151974](https://doi.org/10.5281/zenodo.22151974)